

Case Report I

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1. Framing the decision

Structure the problem by answering the following questions:

■ What do we want? (What is the objective of Recreational Properties?)

The objective of Recreational Properties is to acquire land parcels or properties with recreational parcels, consolidate and lease these parcels to create desirable recreational properties, and maximize returns. Specifically, for the White Mountain Ski Area project, the goal is to secure a lease from the EU Environment Agency, allowing for development and subsequent sale at a high return.

■ What can Anders do? (List the decisions that Anders is facing)

- 1) **Exercise the option:** If Anders exercises the option, he will pay \$10 million to acquire the three parcels of the White Mountain.
 - If lease is obtained
 - Sell the land without development: Anders could sell the land at an estimated price of \$14 million.
 - Develop and sell the land: Anders might invest \$5 million to install infrastructure (access roads, sewers, ski lifts, lodge wing), leading to a potential sale price of \$21.8 million with a 75% chance of a good reputation or \$16.3 million with a 25% chance of a bad reputation.
 - If lease is not obtained: Anders could only sell the land at an estimated price of \$8 million.
- 2) **Abandon the option:** Anders might abandon White Mountain Project option and has the opportunity to participate in another development project with a guaranteed profit of \$1.5 million.

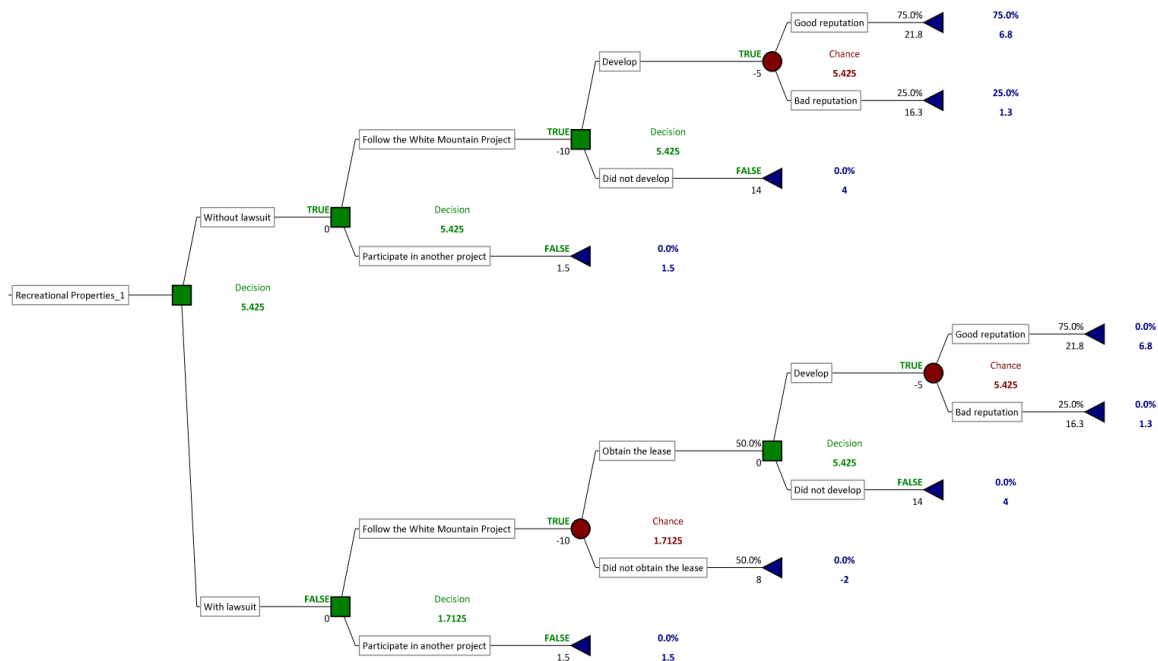
■ What should Anders worry about? (List the uncertainties and how they affect the value of the White Mountain development)

- 1) **Lease Approval:** Anders has a 50% chance of securing the lease from the Environment Agency, which is contingent on the outcome of a lawsuit. If the lease is not approved, the project cannot proceed, and Anders would be limited to selling the land for \$8 million.
- 2) **Reputation of the Developed Ski Resort:** If the lease is approved and the project moves forward, the project's success will depend on its reputation. There is a 75% chance of getting a good reputation and Anders could sell the resort for \$21.8 million,

while there is a 25% chance of receiving a poor reputation, the sale value would be \$16.3 million.

2. The emergency meeting

Anders Forsgren has scheduled an emergency meeting with the board of directors of Recreational Properties:



■ He asks you to find the best course of action under the current circumstances. The decision tree reveals that Anders faces an important choice: whether to continue with the “White Mountain Development Project” or pursue an alternative project that guarantees a profit of \$1.5 million. From an expected value perspective, the overall anticipated return from the White Mountain Project is \$1.7125 million, which is higher than the guaranteed profit from the other project. Therefore, it is preferable to proceed with the White Mountain Project based on the expected value analysis.

■ He would also like to know how the value of the White Mountain development has changed due to the unexpected conservationists’ lawsuit.

The conservationists’ lawsuit has increased the uncertainty of obtaining the lease from 100% to 50%, which has lowered the expected profit value of the White Mountain Project.

- With the lawsuit: The expected profit for the White Mountain Project is \$1.7125 million.
- Without the lawsuit: If we assume there is no lawsuit and a 100% chance of obtaining the lease, we can discard the “did not obtain the lease” path and only consider the outcome of obtaining the lease. In this scenario, the expected profit

value would be \$5.425 million.

Therefore, the unexpected conservationists' lawsuit reduces the project's expected value from \$5.425 million to \$1.7125 million, resulting in a value loss of \$3.7125 million.

- **What would be a reasonable investment if Anders wanted to convince the Environment Agency to grant Recreational Properties the lease, for instance by performing a sophisticated environmental impact report? (Assume the report is positive and would secure the lease for sure.)**

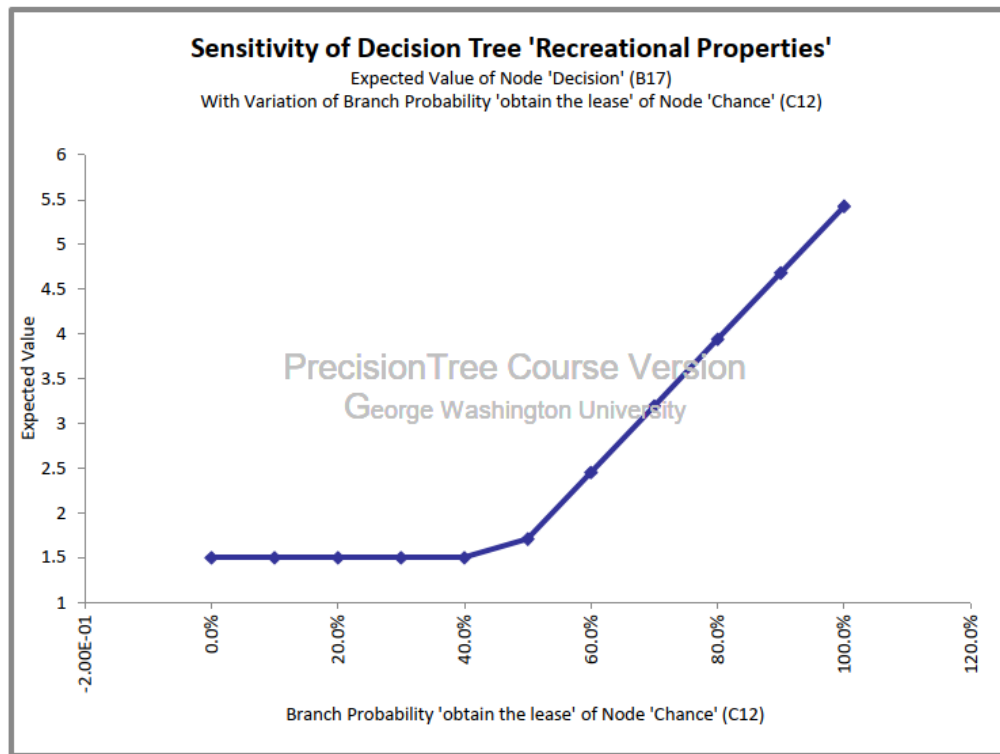
The theoretical investment amount should indeed be less than the difference between the expected value with and without securing the lease. The current project's expected value is 1.7125 but if the lease is guaranteed, the expected value will be 5.425. Reasonable investment should be less than $5.425 - 1.7125 = 3.7125$.

Therefore, the maximum Anders should be willing to invest is 3.7125. If he invests more than this amount, the investment would not yield a favorable return, as the cost would exceed the increase in expected value.

3. Sensitivity Analysis:

After the emergency meeting, Dolores has second thoughts about the robustness of the strategy Anders presented to the board of directions. Especially Anders' estimates of the likelihood of obtaining the lease and of getting a good reputation seemed a bit rough. She thinks a sensitivity analysis of these probabilities is in order.

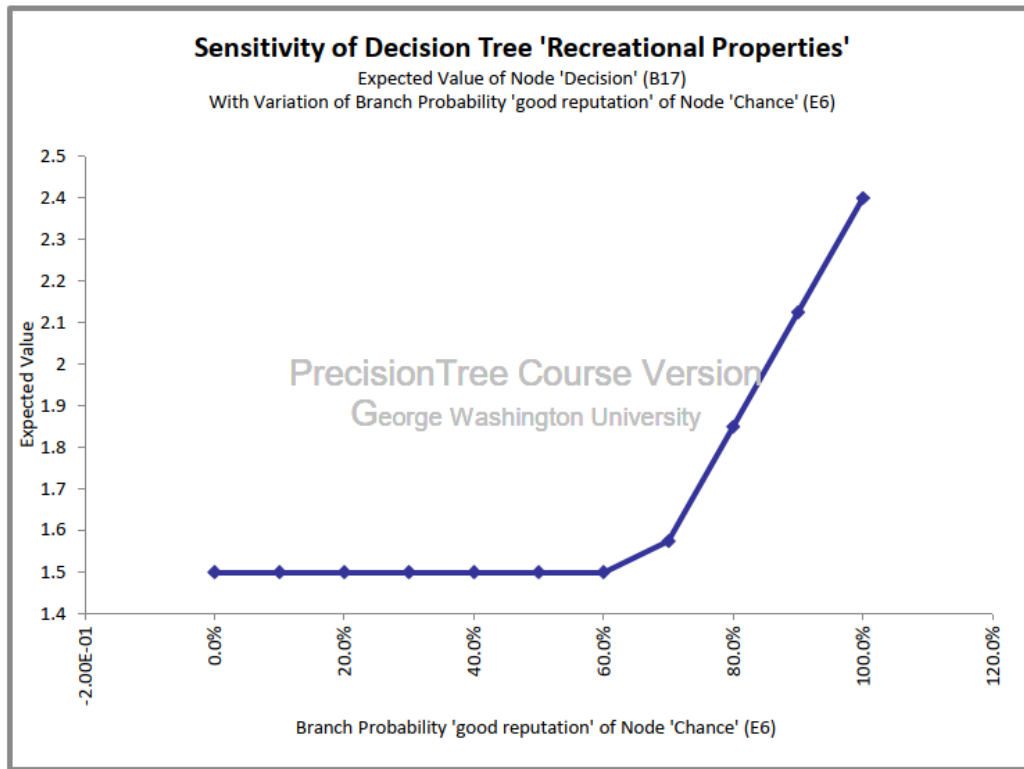
- **How does the recommended strategy change with changes in the estimated probabilities? Can you make a sound recommendation at this point? Justify your answer.**



As we can see from the sensitivity analysis (Expected Value With Variation of Obtaining Lease) graph above, as the probability of obtaining the lease increases, the expected project value rises significantly, indicating that the recommended strategy of proceeding with the White Mountain Project is highly sensitive to the probability of obtaining the lease. Particularly:

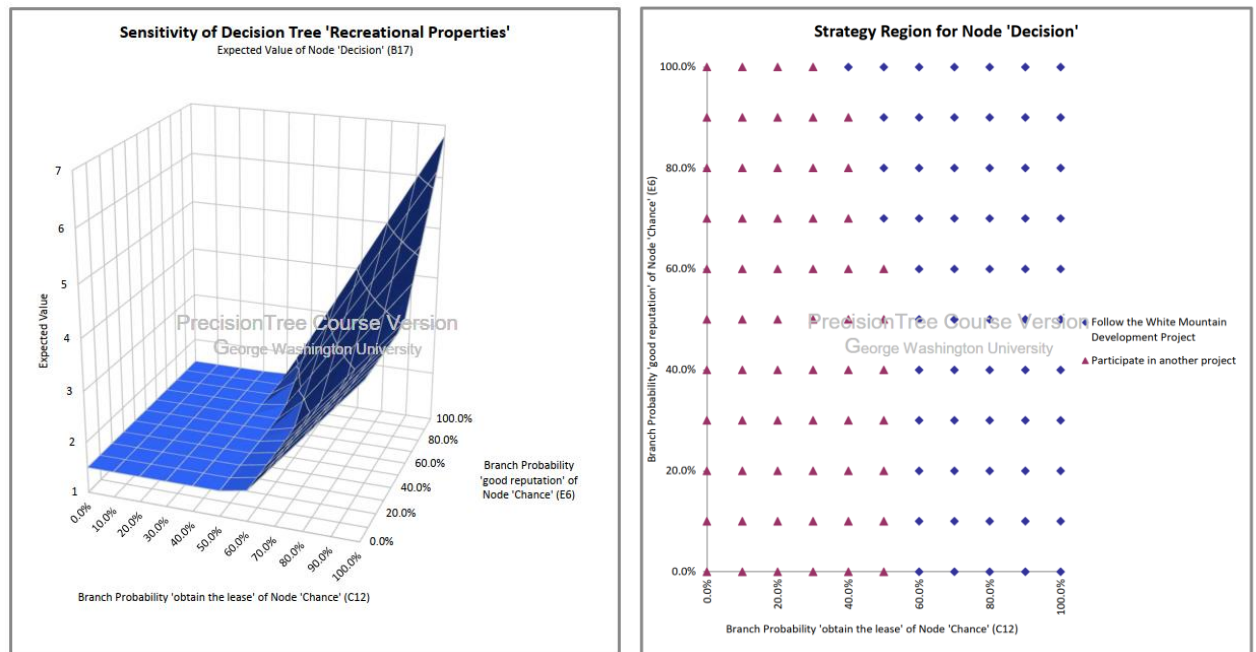
- When the probability is between 0% and 40%, the expected value remains at a relatively low level (close to 1.5), showing that with a low probability of obtaining the lease, the project might have higher risks and might offer minimal return.
- In the 50% to 60% range, the expected project value begins to increase gradually, suggesting that the project's value is not significantly elevated.
- At the probabilities above 70%, the expected project value rises drastically, indicating a significant potential return and making the project much more attractive.

Thus, as the probability of obtaining the lease increases, the recommended strategy's expected return value also increases. When the probability is below 40%, the return on the project does not justify the risk, making an alternative project more prudent. However, at 70% or higher, the White Mountain Project becomes increasingly robust and attractive.



As we can see from the sensitivity analysis (Expected Value With Variation of Good Reputation) graph above, as the probability of getting a good reputation starts to increase, the project's expected value remains constant at first and increases from the probability of 60%. This indicates that at lower probabilities, the project's expected return does not increase significantly, implying a higher risk.

- In the 0% to 60% range, the expected project value stays close to 1.5, similar to the expected value of the alternative project. In this case, proceeding with the White Mountain Project is not a robust recommendation and considering another project would be a safer choice.
- When the probabilities above 60%, the expected project value rises sharply, reaching over 2.0. This indicates with a higher probability of good reputation, the project's expected return becomes greater, making the strategy of proceeding with the White Mountain Project more robust and attractive.



When the probability of obtaining the lease is below 50% and the probability of a good reputation is below 60%, the decision shown on the graph is to **participate in another project**.

When the probability of obtaining the lease exceeds 50%, regardless of the probability of having a good reputation, the decision shifts to **follow the White Mountain Development Project**.

4. Extending the option

Create a new copy of the file “Recreational Properties 2” and call it “Recreational Properties 4” (so that you again have a clean spreadsheet model without too much extra baggage).

After a brainstorming session with her fellow students, Dolores came up with an idea. Perhaps Recreational Properties should negotiate a one-year extension of the options’ maturity date?

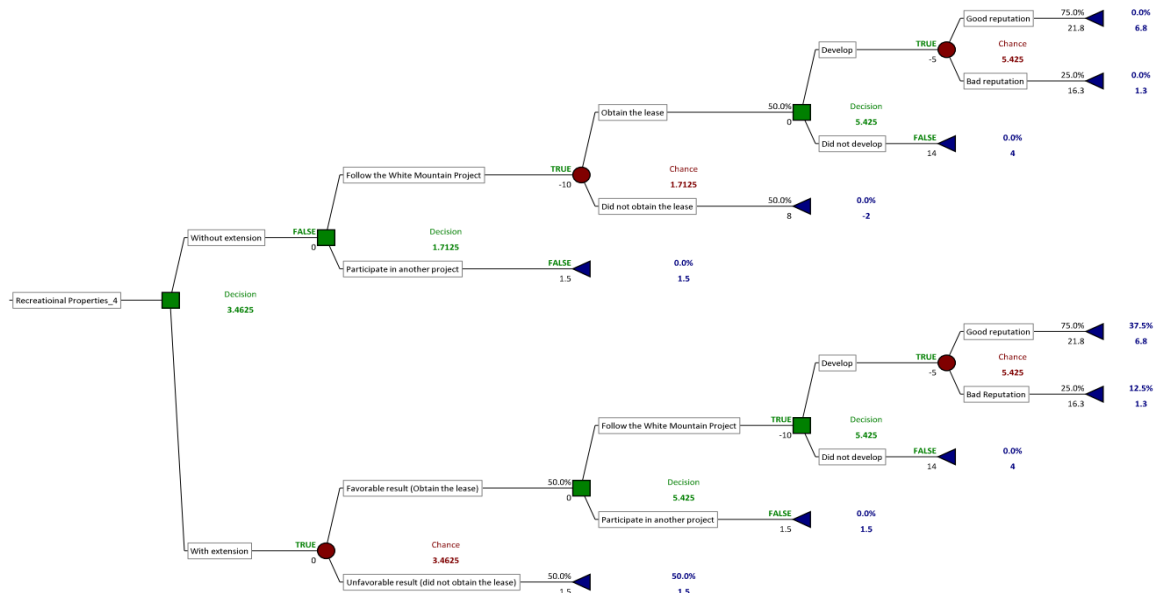
■ Explain why Dolores thinks this might be a good idea.

By extending the option's maturity date by one year, we would be able to decide whether to exercise the purchase option after knowing the final outcome of the lawsuit. If the lawsuit's result is unfavorable, we can choose not to exercise the option and instead participate in another development, which would result in a certain \$1.5 million profit. If the lawsuit's outcome is favorable, we can exercise the purchase option and proceed with the land development, thus avoiding a situation where we exercise the option but do not obtain the lease, which could lead to losses.

When Dolores mentioned her idea to Anders, he replied that he was already thinking about this, but that he was unsure whether this would be a good idea, because a one-year extension, besides being difficult to get, might be very expensive. Therefore, he would like to know before

any negotiations start, how much this would be worth to him?

- How much would a one-year extension on the options be worth? (You can assume that the alternative investment opportunity generating \$1.5 million is also there one year from now.)



Assuming that the alternative investment opportunity generating \$1.5 million is still there, the possible outcomes after knowing the lawsuit's result are:

- 1) If the outcome of the lawsuit is unfavorable, which means that Anders is not able to obtain the lease, the better choice is pursuing the alternative investment with guaranteed profit of \$1.5 million.
- 2) If the outcome of the lawsuit is favorable, which means that Anders is able to obtain the lease, it's better to develop the ski area because the expected development is \$5.425 million ($75\% \times \$6.8 + 25\% \times \1.3).

Given the 50% chance of a favorable outcome, the expected value with an extension is \$3.4625 million ($50\% \times \$5.425 + 50\% \times \1.5). Therefore, the value of a one-year extension is the difference between the expected value profit with an extension is \$ 3.4625 million and the current expected profit without an extension is \$1.7125 million (as calculated previously), which is $\$3.4625 \text{ million} - \$1.7125 \text{ million} = \1.75 million .

- How does this value compare with the loss of value caused by the conservationists' lawsuit (see question 2)? Interpret this result.

The loss of value caused by the conservationists' lawsuit amounts to \$3.7125 million, as the lawsuit introduces uncertainty by reducing the probability of obtaining the lease from 100% to 50%, significantly diminishing the project's expected profitability. However, with a one-year extension, we gain the opportunity to delay the decision until after knowing the

result of the lawsuit. In this situation, Anders can decide whether to spend \$10 million to acquire the three parcels of White Mountain with an expected profit of \$5.425 million or pursue another project with a guaranteed profit of \$1.5 million. Based on prior calculations, the expected profit with the extension is \$3.4625 million. Therefore, spending \$1.75 million to extend the decision represents a strategic move to mitigate risk by waiting for the lawsuit outcome, which remains significantly less than the \$3.7125 million reduction caused by the lawsuit.

5. A first meeting with the landowners

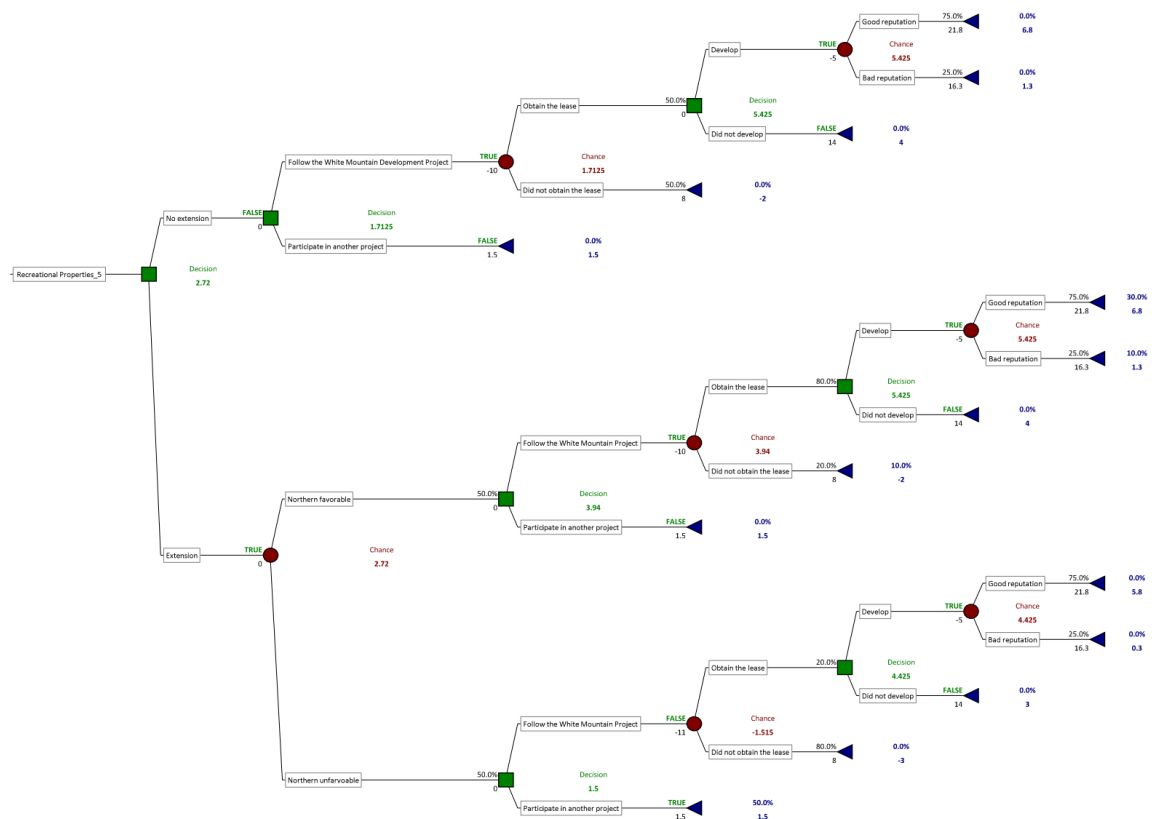
Create a new copy of the file called “Recreational Properties 5”. After a first meeting with the three property owners, it seemed that the owners were not inclined to sell Anders the extension of the options. Anders thought he could negotiate at most a six-month extension of the options’ maturity date, until December 1, 2002. This was the best he could hope for.

Besides the White Mountain development project, there was a Northern Development project going on in the same region, on which Anders was not involved. Anders was sure the court decision regarding the northern development would be made by late August. However, he knew that no matter which way the court decided, a series of public hearings would have to be held before the Environment Agency would decide on the White Mountain lease. These hearings would not start until after the northern development court decision was reached in late August, and he was confident the decision on the White Mountain lease would be reached sometime between early February and late March 2003.

Anders thought there was a 50% chance that the law court would decide to halt the northern development, as the case for and against seemed equally strong. However, a law court decision to halt the northern development would not necessarily block the White Mountain development. If the northern development suit was settled unfavorably, he thought the chance that he would obtain a lease for the White Mountain development would be 20%. On the other hand, if the court case turned out to be favorable, Anders considered he would be in a very strong position to obtain a lease, at least an 80% chance.

Anders was meeting with the property owners the next day, and was trying to determine a reasonable price for the six-month extension of the options’ maturity date:

- **Determine a reasonable price for the six-month extension.**



The reasonable price for the six-month extension is the difference between the expected value with the extension and the expected value without the extension, which is \$1.0075 million ($\$2.72 - \$1.7125 = \1.0075).

- How does this price compare with the price of a one-year extension? Interpret this result.

1) The benefits and cost of each extension option

- **Six-Month Extension**

Expected Value = \$2.72 million

Expected Value increased = Cost of extension = $\$2.72 - \$1.7125 = \$1.0075$

- **One-Year Extension**

Expected Value = \$3.4625 million

Expected Value increased = Cost of extension = $\$3.4625 - \$1.7125 = \$1.75$

2) Information value of each extension option

- **Six-Month Extension**

The six-month extension can provide some reference information on the probability of obtaining the lease based on the outcome of the Northern Development court decision, but still leaves uncertainty unresolved.

- **One-Year Extension**

The one-year extension provides almost complete information, including the Northern Development court decision, public hearings, and even the Environment

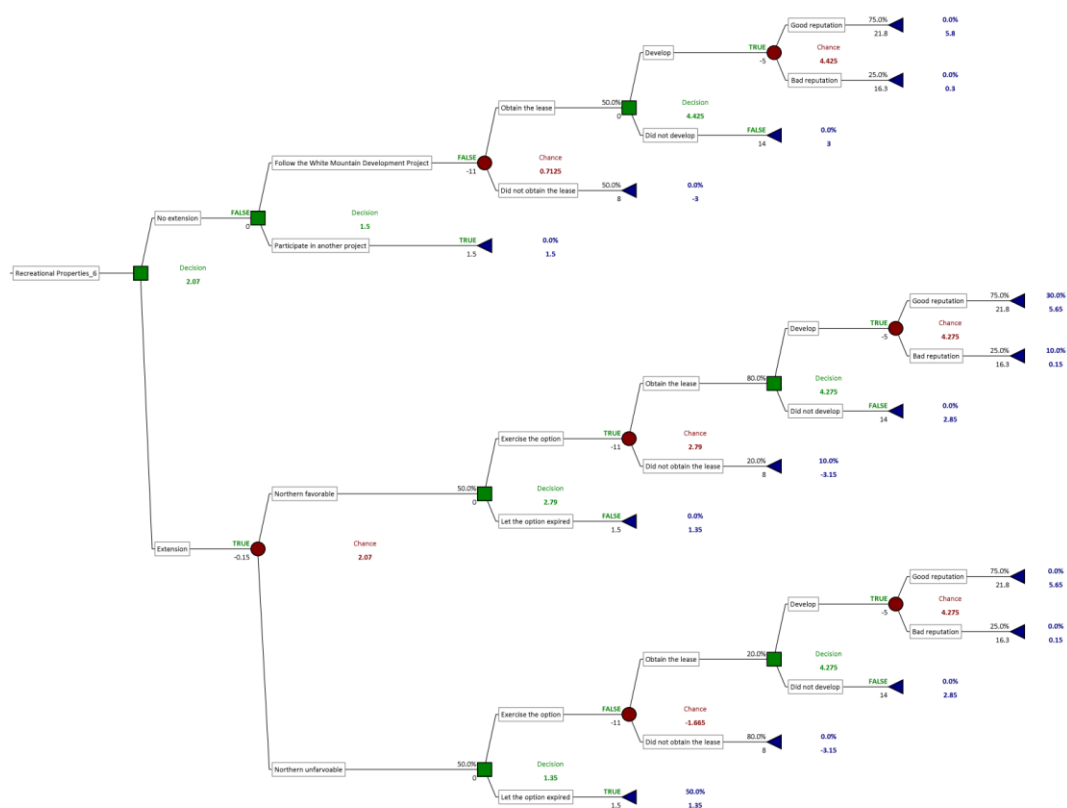
Agency's lease decision, which eliminates all major uncertainties and allows for informed decision-making.

6. The final negotiation

Create a new copy of the file called "Recreational Properties 6".

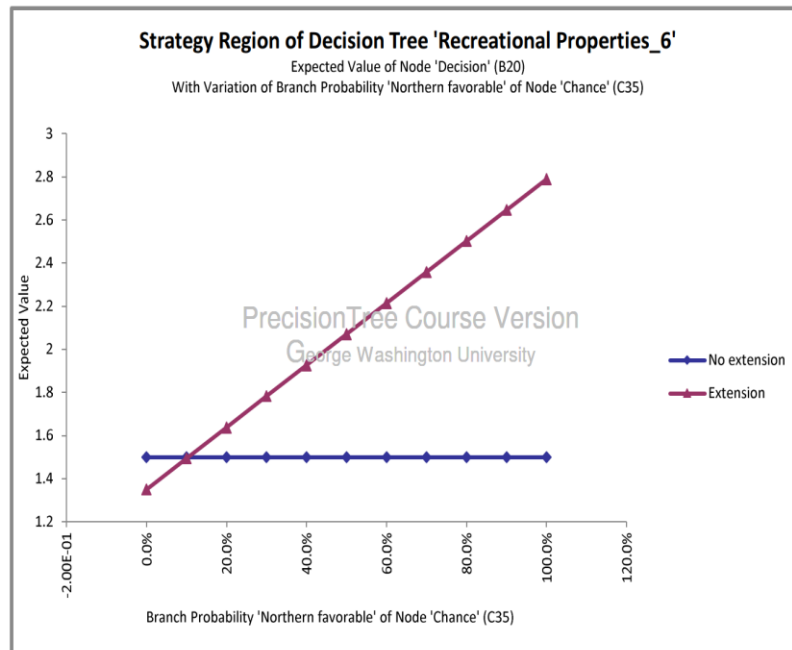
The three owners propose to Anders the following deal: the price of the extension itself would be around \$150,000, and the options' strike price would increase to \$11 million.

■ Should Anders accept this offer?



Since the options' strike price is \$ 11 million, the expected profit of no extension will become \$ 1.5 million (choosing another project) and the expected profit in the extension scenario will be \$ 2.07 million. So, Anders should accept the offer because the expected profit in the extension scenario is higher than the expected profit of no extension scenario.

■ Would you buy the six-month extension if the probability of a favorable court ruling were different from 50%? Interpret this result.



Strategy Region Data						
	Input		No extension		Extension	
	Value	Change (%)	Value	Change (%)	Value	Change (%)
#1	0.0%	-100.00%	1.5	-27.54%	1.35	-34.78%
#2	10.0%	-80.00%	1.5	-27.54%	1.494	-27.83%
#3	20.0%	-60.00%	1.5	-27.54%	1.638	-20.87%
#4	30.0%	-40.00%	1.5	-27.54%	1.782	-13.91%
#5	40.0%	-20.00%	1.5	-27.54%	1.926	-6.96%
#6	50.0%	0.00%	1.5	-27.54%	2.07	0.00%
#7	60.0%	20.00%	1.5	-27.54%	2.214	6.96%
#8	70.0%	40.00%	1.5	-27.54%	2.358	13.91%
#9	80.0%	60.00%	1.5	-27.54%	2.502	20.87%
#10	90.0%	80.00%	1.5	-27.54%	2.646	27.83%
#11	100.0%	100.00%	1.5	-27.54%	2.79	34.78%

After conducting a sensitivity analysis, we find that if the probability of a favorable court outcome falls below approximately 10%, it is more beneficial to choose not to extend, as the expected value of no extension would be higher. Conversely, when the probability exceeds 10%, the expected value of extending becomes greater than not extending. Therefore, in cases where the probability is higher than 10%, opting for an extension would be the preferable decision.